

Software Requirements Specification

(SRS)

Project Name	HEALTH AI Co-Creation & Innovation Platform		
Course	SENG384		
Group No / Name	WebStar		
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Date	Version	Change Description	Author
16/04/2026	1.1	Initial version	Sena Nur Pekgöz Ceren Kızılay

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1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the HEALTH AI Co-Creation & Innovation Platform. This SRS serves as the primary guide for the development team and stakeholders to ensure all system objectives are met.

1.2 Scope

The platform enables structured partner discovery between healthcare professionals and engineers to foster multidisciplinary innovation. It provides a secure matchmaking infrastructure but does not provide financial transactions, contract management, medical advice, or file-sharing services.

1.3 Definitions and Abbreviations

SRS: Software Requirements Specification

RBAC: Role-Based Access Control

GDPR: General Data Protection Regulation

NDA: Non-Disclosure Agreement

WCAG: Web Content Accessibility Guidelines

1.4 Intended Audience

This document is intended for the WebStar development team, course instructors, and the Erasmus+ HEALTH-AI project evaluators.

2. Overall Description

2.1 Product Perspective

Multidisciplinary innovation in healthcare currently relies on coincidence or personal networks, which is inefficient. HEALTH AI addresses this gap by providing a structured, announcement-based discovery platform that eliminates randomness in interdisciplinary collaboration.

2.2 User Roles

Describe the user roles in the system and the key responsibilities of each.

Role	Description	Key Permissions
Engineer	Technical expert seeking clinical domain knowledge.	Create posts, browse medical ads, request meetings.
Healthcare Professional	Medical expert with innovation ideas lacking technical infrastructure.	Create posts, browse technical ads, request meetings.

Admin	System overseer responsible for platform integrity.	Manage users, audit logs, remove inappropriate content.
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2.3 Assumptions and Dependencies

- Users are assumed to have a valid institutional .edu email address.
- The system depends on external meeting tools (Zoom/Teams) for the actual collaboration

2.4 Constraints

- **No Data Storage:** The system shall not store patient data or confidential technical files to minimize legal risk.
- **Communication:** The platform only facilitates the "first contact" and meeting scheduling

3. Functional Requirements

3.1 User Registration & Access Control

ID	Requirement Description	Priority	Source
FR-01	The system shall only allow registration with institutional .edu email addresses.	High	Brief 4.1
FR-02	The system shall implement an email verification mechanism.	High	Brief 4.1
FR-03	The user shall be able to select a role during registration (Engineer / Healthcare Professional / Admin).	High	Brief 4.1
FR-04	[Add your own requirements]	[...]	[...]

3.2 Post Management

ID	Requirement Description	Priority	Source
FR-10	Users shall create structured posts defining domain, stage, and expertise needed.	High	Brief 4.2
FR-11	Posts must track status: Draft, Active, Meeting Scheduled, Partner Found, Expired.	High	Brief 4.2
FR-12	The system must strictly prohibit file uploads for documentation or patient data.	High	Brief 4.2

Carefully review section 4.2 of the brief to cover all post management requirements. Do not forget the post lifecycle states (Draft, Active, Meeting Scheduled, Partner Found, Expired).

3.3 Search & Matching

ID	Requirement Description	Priority	Source
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FR-20	[Write requirement]	[...]	Brief 4.3
FR-21	[Write requirement]	[...]	Brief 4.3

3.4 Meeting Request Workflow

ID	Requirement Description	Priority	Source
FR-30	[Write requirement]	[...]	Brief 4.4
FR-31	[Write requirement]	[...]	Brief 4.4

3.5 Administrative Dashboard

ID	Requirement Description	Priority	Source
FR-40	[Write requirement]	[...]	Brief 4.5
FR-41	[Write requirement]	[...]	Brief 4.5

3.6 Activity Logging & Audit Trail

ID	Requirement Description	Priority	Source
FR-50	[Write requirement]	[...]	Brief 4.6
FR-51	[Write requirement]	[...]	Brief 4.6

4. Non-Functional Requirements

ID	Requirement	Metric / Target	Category
NFR-01	Search results shall be returned within a specified time.	< 1.5 seconds	Performance
NFR-02	Page load time shall be limited.	< 3 seconds	Performance
NFR-03	Learnability	Purpose understandable within 20 seconds .	Usability
NFR-04	Data Privacy	Support for "Right to delete" and "Right to export".	GDPR
NFR-05	Logging	Logs must be tamper-resistant and retained for 24 months .	Audit

5. Use Cases

5.1 UC-01: Engineer Creates a Post

Name	UC-01: Engineer Creates a Post
Actor(s)	Engineer

Precondition	The engineer must be logged in.
Main Flow	1. The engineer clicks the “New Post” button. 2. Fills in the required fields (domain, description, required expertise, etc.). 3. Clicks the “Publish” button. 4. The system publishes the post with “Active” status.
Postcondition	The post is saved and visible to other users.
Alternative Flow	2a. If a required field is left blank, the system displays an error message.

5.2 UC-02: Requesting a Meeting

Name	UC-02: Requesting a Meeting
Actor(s)	Engineer or Healthcare Professional
Precondition	User is logged in.
Main Flow	1. User browses posts. 2. Selects an interesting project. 3. Accepts the NDA tooltip/pop-up. 4. Proposes a time slot. 5. Sends request.
Postcondition	
Alternative Flow	

5.3 UC-03: [Use Case Name]

Name	UC-03: [Use Case Name]
Actor(s)	[Actor]
Precondition	[Write]
Main Flow	[Write steps]
Postcondition	[Write]
Alternative Flow	[Write]

Add more use cases as needed (UC-04, UC-05, ...). Consider scenarios such as meeting request, admin removing a post, marking partner found.

6. Data Model

Entity	Key Fields	Relationships
User	ID, Email (.edu), Name, Role.	Creates N Posts, Sends N MeetingRequests.
Post	ID, Title, Domain, Stage, Status, CreatorID.	Belongs to 1 User, Has N MeetingRequests.
MeetingRequest	ID, SenderID, PostID, Status, TimeSlots.	Connects 2 Users via 1 Post.
ActivityLog	Timestamp, UserID, ActionType, IP.	Tracks 1 User action.

7. Interface Requirements

7.1 User Interface (UI)

- **LandingPage:** Quick value proposition and CTA.
- **PostsPage:** Searchable grid with filters for domain and city.
- **PostEditorPage:** A 3-step form to ensure fast creation (< 3 min).
- **AdminPage:** Dashboard for role-based user and log management.

7.2 External System Interfaces

- **Email Gateway:** For SMTP-based email verification.
- **Calendar/External Links:** Generation of Zoom/Teams links (metadata only).

8. Requirements Traceability Matrix

Show the source of each requirement in the project brief and which use case it relates to. Ensure all requirements are covered.

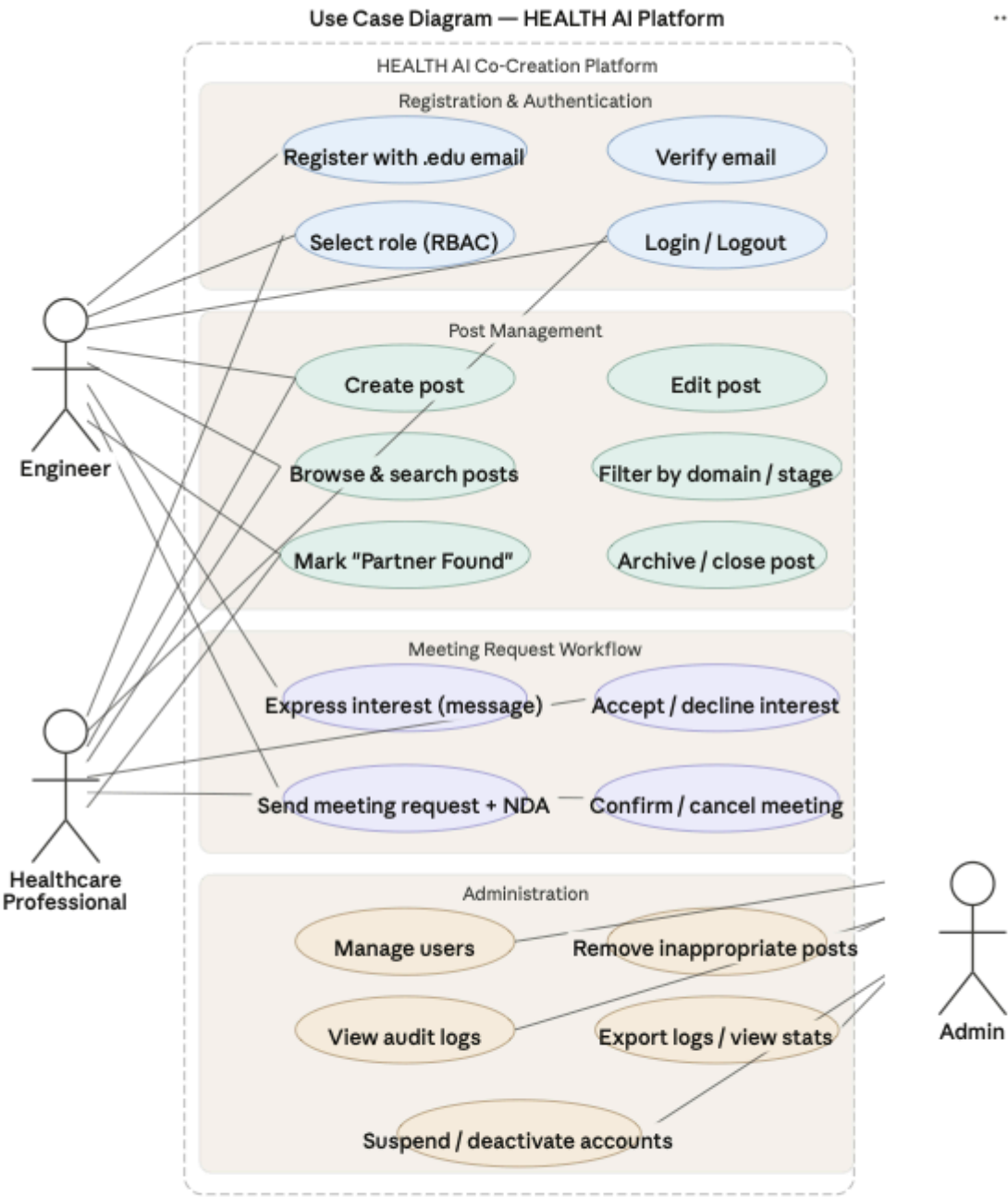
Req. ID	Requirement Summary	Source (Brief)	Related Use Case
FR-01	.edu email restriction	Brief 4.1	UC-01, UC-02

9. Appendices

• Use Case Diagram

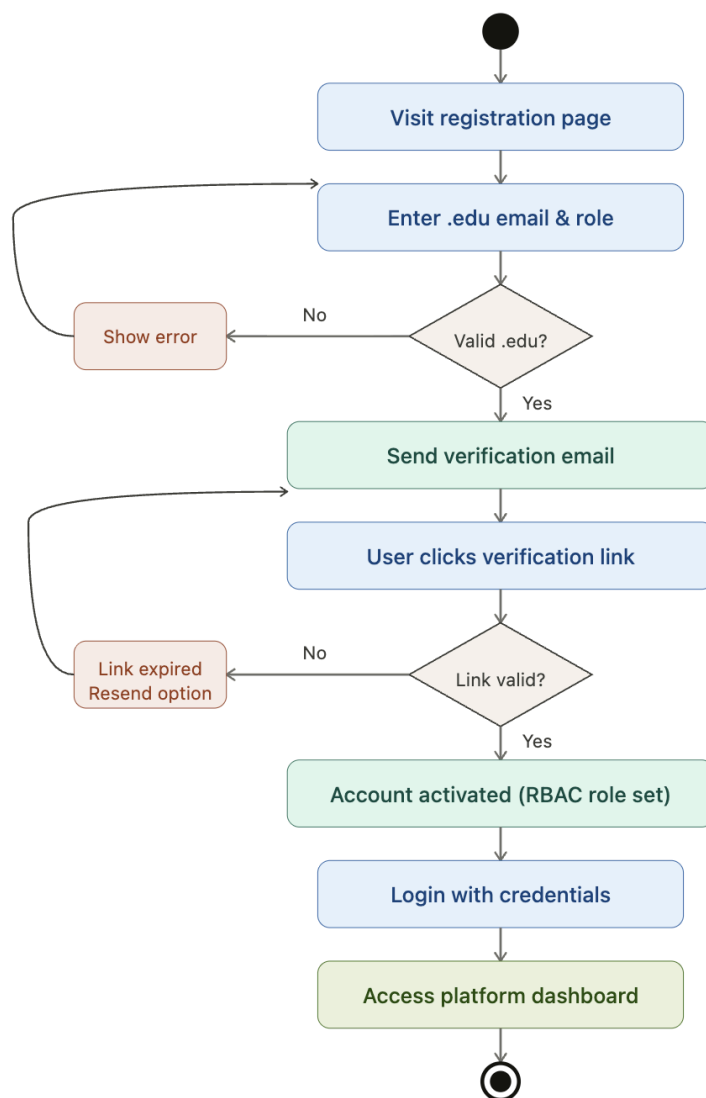
Visualize Widget

16.04.2026 17:25

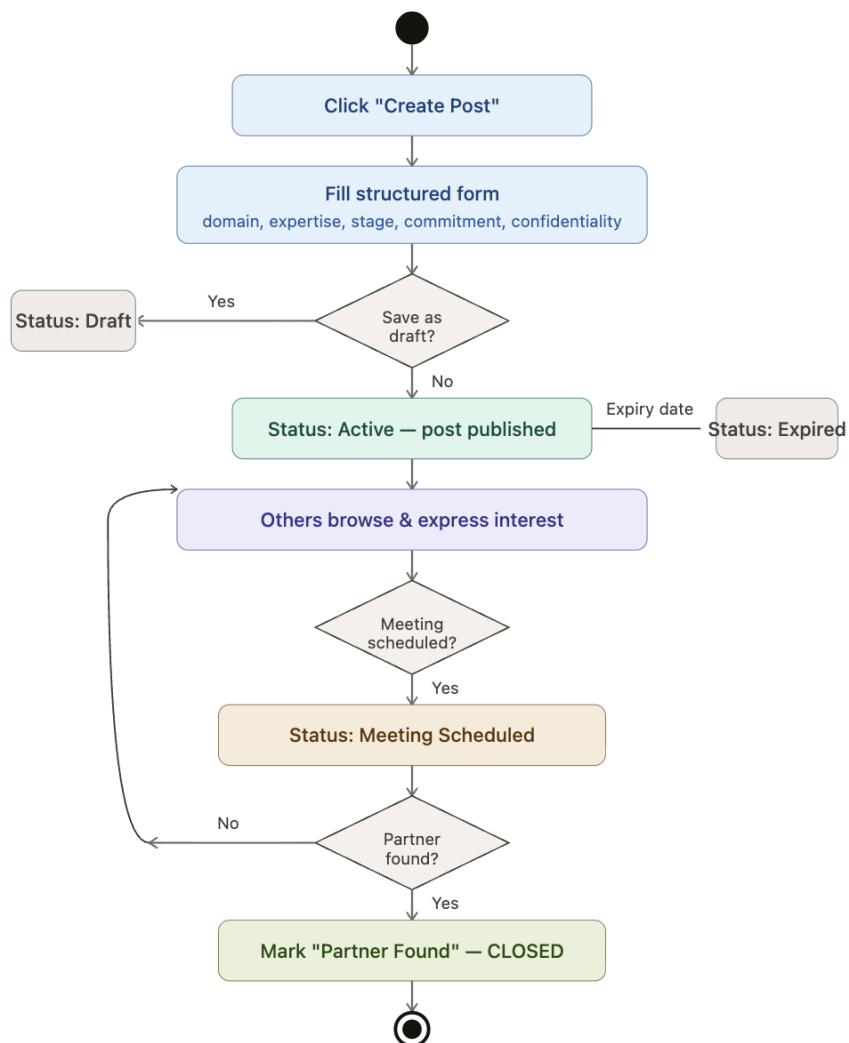


• Activity Diagrams

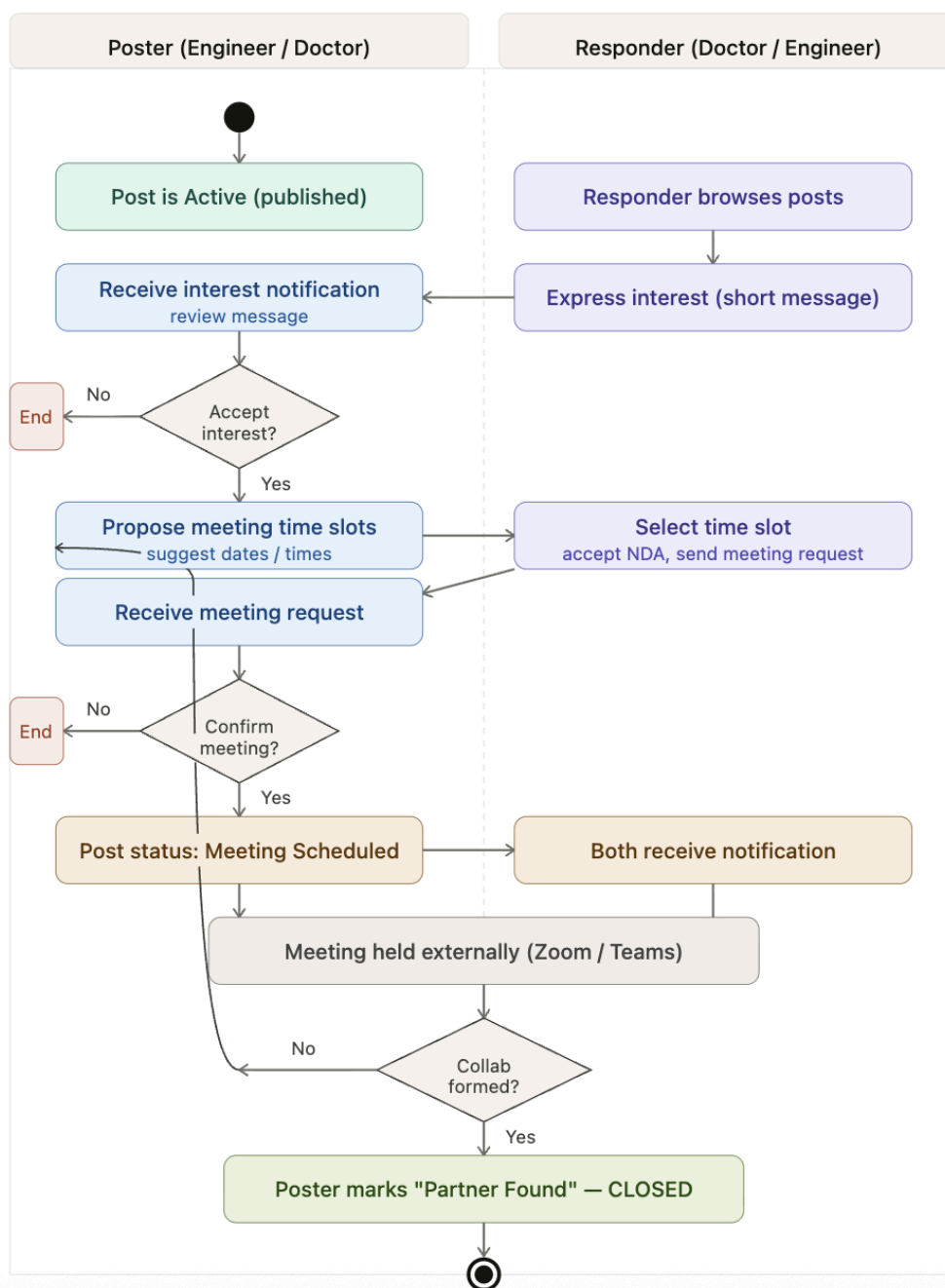
Activity Diagram 1 — User Registration & Login



Activity Diagram 2 — Post Creation & Lifecycle

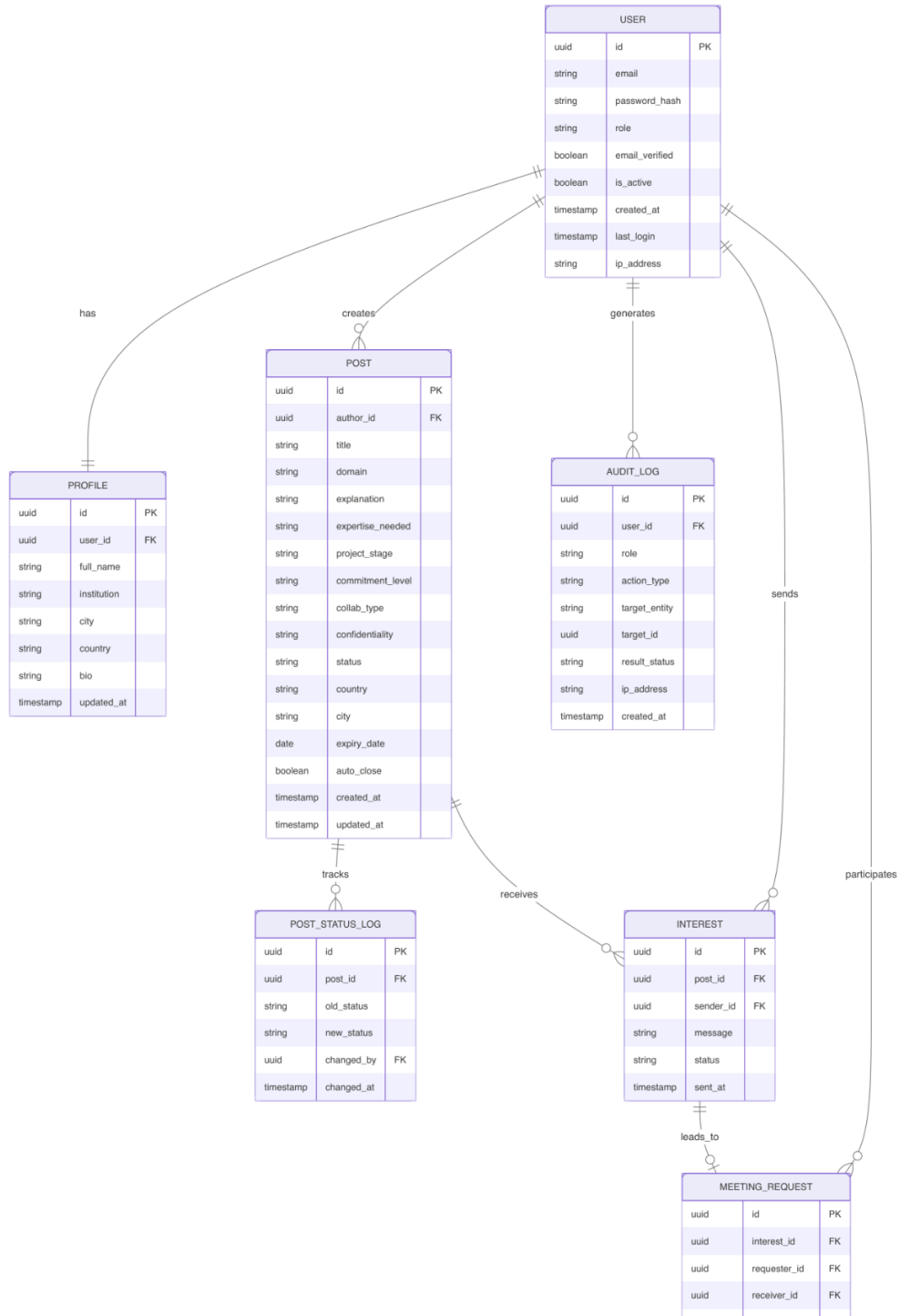


Activity Diagram 3 – Meeting Request Workflow

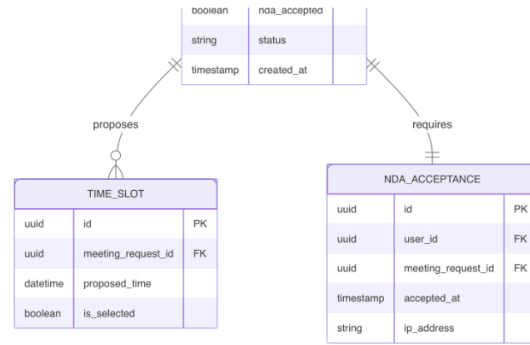


• ER Diagram

16.04.2026 17:51



16.04.2026 17:51



- Wireframes

Screen 1 - Landing Page

16.04.2026 17:59

HEALTH AI

[Browse posts](#)[How it works](#)[Sign in](#)[Join platform](#)

GDPR-compliant · .edu only · NDA-protected

Connect healthcare professionals with engineers

A secure matchmaking platform for structured interdisciplinary collaboration in health tech.

Post your idea, find your partner.

[Post an announcement](#)[Browse active posts](#)

142

Active posts

38

Matches formed

9

Countries

HOW IT WORKS



Post an announcement

Describe your idea, needed expertise, and commitment level – without revealing sensitive details.



Find your match

Browse posts filtered by domain, city, and expertise. Send an interest message to start the conversation.



Schedule a meeting

Propose time slots, accept NDA, and meet externally on Zoom or Teams. Platform closes cleanly.

RECENT POSTS

16.04.2026 17:59

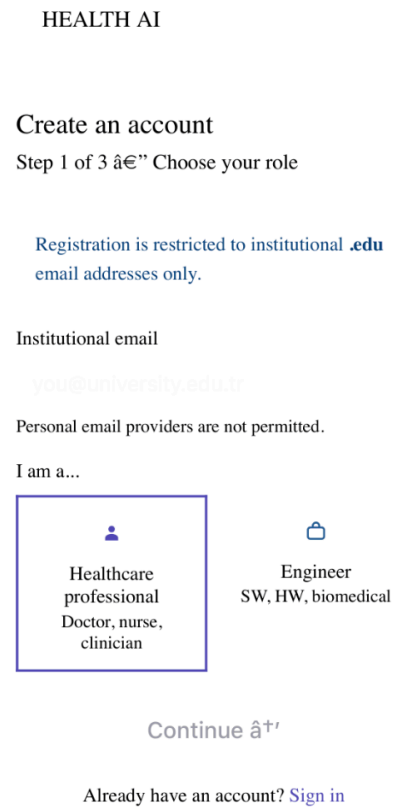
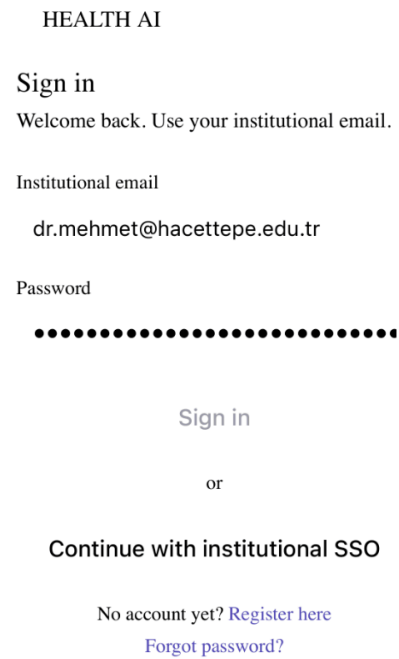
DR	ML model for early sepsis detection in ICU Intensive care Â· Istanbul Â· Seeking ML engineer	Healthcare
EK	Wearable ECG analysis â€” clinical validation needed Cardiology imaging Â· Ankara Â· Prototype stage	Engineer
AS	Surgical workflow optimization using computer vision Surgical robotics Â· Berlin Â· Concept validation	Healthcare

HEALTH AI Co-Creation Platform Â· GDPR compliant Â· .edu registration only

Privacy policy Â· Terms

Screen 2 - Auth Page

16.04.2026 17:59



Screen 3 - User Profile Page

16.04.2026 18:00

HEALTH AI Browse My posts Profile ZA



Dr. Zeynep Arslan

Intensive Care Specialist Â Istanbul Technical University

Healthcare professional Istanbul, TR Email verified

Edit profile

ABOUT

ICU specialist with 8 years of clinical experience at ITU Hospital. Interested in applying machine learning to early-warning systems for sepsis and hemodynamic instability. Looking for ML engineers with healthcare domain experience.

PROFILE DETAILS

Institution	Role
Istanbul Technical University	Healthcare professional
City	Country
Istanbul	Turkey
Email	Member since
dr.zeynep@itu.edu.tr	March 2025

EXPERTISE DOMAINS

- Intensive care
- Sepsis detection
- Hemodynamics
- Clinical workflows
- Patient monitoring
- Ethical approval

ACTIVITY

3

Posts created

7

Interests received

1

Match formed

MY POSTS

ML model for early sepsis detection in ICU

Meeting scheduled

Intensive care

Posted 12 days ago

5 interests

16.04.2026 18:00

NLP-based clinical note summarization tool
Clinical NLP Posted 3 days ago 2 interests

Active

Wearable alert system for at-risk patients
Patient monitoring Closed 2 months ago

Partner found

ACCOUNT SETTINGS

Change password

Update

Export my data

Download

Delete account

Delete

Screen 4 - Creating Post Page

16.04.2026 18:00

HEALTH AI

 Back to posts

ZA

New announcement

Describe your project without revealing sensitive IP details.

Your announcement will be visible to all verified platform users. Confidential technical details should only be shared in your first meeting.

BASIC INFORMATION

Title

ML model for early sepsis detection in ICU

Short, descriptive. Avoid revealing specific methods.

Medical / engineering domain

Intensive care

Expertise needed

Machine learning engineer

Short explanation (public pitch)

Seeking an ML engineer to co-develop a real-time early warning system for sepsis in ICU patients. Clinical data access available through institutional channels. Looking for expertise in time-series modeling and clinical AI.


 214 / 500

PROJECT DETAILS

Project stage

Idea

Concept validation

Prototype developed

Pilot testing

Pre-deployment

Collaboration type

Research partner

Commitment level

Medium " regular meetings

Country

Turkey

City

Istanbul

VISIBILITY & EXPIRY

Confidentiality level

Public short pitch

Summary visible to all users

Details in meeting only

Only title shown publicly

16.04.2026 18:00

Expiry date

01.09.2025

Post closes automatically on this date.

☒ Auto-close when partner found

Save as draft

Publish announcement

Screen 5 - Meeting Request Page

16.04.2026 18:00

HEALTH AI

Back to post

EK

Interest sent 2 Meeting request 3 Confirmed

RESPONDING TO POST

ZA ML model for early sepsis detection in ICU
 Dr. Zeynep Arslan Istanbul Healthcare Concept validation

YOUR INTRODUCTION

Message to Dr. Arslan

Hi Dr. Arslan, I'm a senior ML engineer specializing in time-series anomaly detection at Boğaziçi University. I've worked on clinical NLP projects and am very interested in collaborating on early-warning systems for ICU patients. I believe my experience with LSTM and transformer models could be directly applicable here.

312 / 600

Keep it professional. Do not include external links or contact details.

PROPOSE MEETING TIMES

Meetings are conducted externally (Zoom, Teams). The platform does not store recordings or notes.

Date	Time (your timezone + UTC+3)	
05.08.2025	14:00	—
06.08.2025	10:30	—
07.08.2025	16:00	—

+ Add another time slot

NON-DISCLOSURE AGREEMENT

Mutual NDA + HEALTH AI Platform

By sending this meeting request, both parties agree to maintain confidentiality regarding any ideas, technical concepts, business plans, or clinical data discussed during the meeting facilitated through this platform.

16.04.2026 18:00

Neither party shall disclose, share, or use the other party's confidential information for any purpose other than evaluating a potential collaboration. This agreement is effective from the moment the meeting request is

☒ I have read and agree to the mutual NDA. I understand that any information shared in the meeting is confidential and may not be disclosed to third parties.

[Cancel](#)[Send meeting request â†’](#)